Accessing State Graphics

__Accessing State Graphics

SimTalk provides the *attributes* listed in the table of contents to the left for **state graphics** of the material flow and of the fluid objects.

See also

Edit 3D Properties [dialog] > **States**

_3D.StatesForColoring [SimTalk]

Sets which states are considered for state coloring of the object designated by <Path>.

Remarks

Assigning the attribute deactivates graphic inheritance.

Type

Attribute

Syntax

```
<Path>._3D.StatesForColoring:string[]
```

Assignment Value

You can assign an *array* of data type *string*.

State of the object	Color
Unplanned	light blue
Paused	blue
Failed	red
Stopped	pink
Powering up/down	purple
Off	dark gray
Standby	light gray
Working	green
Blocked	yellow
Setting-Up	brown
Entrance closed	cyan
Waiting	orange
In execution (*Method*)	dark green
Suspended (*Method*)	purple
Encrypted (*Method*)	gray

State of the object	Color
Syntax error (*Method*)	light red
Program inherited (*Method*)	turquoise

Note

You can only assign the states that apply to the respective object.

Example

```
MyParallelStation._3D.StatesForColoring := ["Failed", "Stopped",
"Unplanned", "Paused", "Working", "Blocked", "Setting-up"]
```

See also

States > **Orientation [state graphic]**

_3D.StatesOrientation [SimTalk]

Sets the orientation and the color of the state graphics of the object designated by <Path>.

Remarks

When you set the attribute, *Plant Simulation* deletes existing state graphics or re-creates them with the orientation you set. *Plant Simulation* also resets the transformation (position and scaling) of the state graphic.

Assigning the attribute deactivates graphic inheritance.

Type

Attribute

Syntax

```
<Path>._3D.StatesOrientation:string
```

Assignment Value

You can assign a value of data type *string*.

You can specify `"(Off)"`, `"Horizontal"`, `"Vertical"`, or `"Color"`.

Example

```
MyStation._3D.StatesOrientation := "Horizontal"
```

See also

States > **Orientation [state graphic]**

_3D.StatesPosition [SimTalk]

Sets the reference point of the state graphic of the object designated by <Path>.

Remarks

Assigning the position moves all state graphics according to these values.

Assigning the attribute deactivates graphic inheritance.

Type

Attribute

Syntax

```
<Path>._3D.StatesPosition:length[]
```

Assignment Value

You can assign an *array* of data type *length*.

Example

```
MyStation._3D.StatesPosition := [-1, 1, 1]
```

See also

States > **Position [state graphic]**

_3D.Position [SimTalk]

Position [SimTalk] - graphic

_3D.StatesScale [SimTalk]

Sets the scaling of the state graphic of the object designated by <Path>.

Remarks

Scaling changes the size and the position of all state graphics.

Assigning the attribute deactivates graphic inheritance.

Type

Attribute

Syntax

```
<Path>._3D.StatesScale:real/real[3]
```

Assignment Value

You can assign:

- A value of data type *real* to use uniform scaling.
- An *array* of data type *real* with three values to use different scaling factors for the three axes.

Return Value

The return value is always an *array* with three values.

Example

```
MyStation._3D.StatesScale := 1          // uniform scaling
MyStation._3D.StatesScale := [1, 4, 9]  // different scaling factors for
the x-axis, y-axis and z-axis
var a := MyStation._3D.StatesScale      // 'a' designates an array
```

See also

States > **Scale [state graphic]**

_3D.Scale [SimTalk]

Scale [SimTalk] - graphic

X [SimTalk] - array

Y [SimTalk] - array

Z [SimTalk] - array

_3D.StatesScaleWithObject [SimTalk]

Scales the state graphic as well, when you scale the object designated by <Path> (`true`) or does not scale the state graphic (`false`).

Remarks

To not scale the state graphic as well, when you scale the object itself, specify `false`. *Plant Simulation* still scales the position of the state graphic though.

Assigning the attribute deactivates graphic inheritance.

Type

Attribute

Syntax

```
<Path>._3D.StatesScaleWithObject:boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MyStation._3D.StatesScaleWithObject := true
```

See also

States > **Scale With Object [check box]**

Accessing Point Clouds

__Accessing Point Clouds

SimTalk provides the *attributes* listed in the table of contents to the left for point clouds of the *Frame*.

See also

Edit 3D Properties [dialog] > **Point Cloud**

_3D.PointCloudPath [SimTalk]

Sets the file path to the point cloud you want to use in the *Frame* designated by <Path>.

Remarks

A relative path refers to the folder in which your simulation model is located.

Type

Attribute

Syntax

```
<Path>._3D.PointCloudPath:string
```

Assignment Value

You can assign a value of data type *string*.

Example

```
.Models.MyFrame._3D.PointCloudPath := "C:\Program Files\Siemens\Plant
Simulation 2606\3D\Siemens_Stand_Layers rotate 180degree.pod"
```

See also

Edit 3D Properties [dialog] > **Point Cloud** > **File Path**

_3D.PointCloudPosition [SimTalk]

Sets the position of the point cloud in the *Frame* designated by <Path>.

Type

Attribute

Syntax

```
<Path>._3D.PointCloudPosition:array
```

Assignment Value

You can assign a value of data type *array*.

The values designate the **X**-position, the **Y**-position, and the **Z**-position of the point cloud in the simulation model.

Example

```
.Models.MyFrame._3D.PointCloudPosition := [2, 3, 0]
var a := .Models.MyFrame._3D.PointCloudPosition
```

See also

Edit 3D Properties [dialog] > **Point Cloud** > **Position [point cloud]**

_3D.PointCloudRotation [SimTalk]

Sets the rotation, i.e., the rotation angle and the rotation axis of the point cloud in the *Frame* designated by <Path>.

Remarks

Plant Simulation interprets the rotation angle as a rotation around the negative z-axis.

Type

Attribute

Syntax

```
<Path>._3D.PointCloudRotation:any
```

Assignment Value

You can specify:

- A number which defines the rotation around the negative z-axis at the position at which you inserted the object.
- An *array* with four values. The first value designates the angle, the remaining three values designate the components of the rotation axis.

Return Value

When querying the attribute, 3D returns an *array* with four items.

Example

```
.Models.MyFrame._3D.PointCloudRotation := 30             // rotation by 30
degrees
.Models.MyFrame._3D.PointCloudRotation := [45, 0, 1, 0]  // rotates by 45
degrees
                                                         // around the y
axis
```

See also

Edit 3D Properties [dialog] > **Point Cloud** > **Rotation [point cloud]**

Accessing Video Recording Settings

_Accessing Video Recording Settings

SimTalk provides the *functions* listed in the table of contents to the left for accessing and recording a video.

F3DfinishVideo [SimTalk]

Finishes the currently running video recording.

Remarks

If you started the video recording with the dialog **Start Simulation Recording** or the function **F3DrecordSimulationVideo [SimTalk]**, finishing the video recording will stop the simulation.

Type

Function

Syntax

```
F3DfinishVideo
```

Example

```
F3DfinishVideo
```

See also

Video Ribbon Tab > **Finish Recording**

stop [SimTalk] - EventController

F3DisRecordingAVideo [SimTalk]

Returns if a video is recorded at the moment or not.

Type

Function

Syntax

```
F3DisRecordingAVideo -> boolean
```

Example

```
-- If a video is currently being recorded we set the camera to different
positions
if F3DisRecordingAVideo
    -- First we want to show the model from a position we generated
    -- with the ribbon command 3D > View > Marks > Generate SimTalk Code...
    F3DconfigureView([1.043, -26.821, 13.465], 58.500, -17.565, .Models.Model)
    wait 15
    -- Now we play the camera path Ani1
    wait _3D.CameraAnimations.Ani1.play
    wait 5
    -- We set the camera to another position and direction
    F3DconfigureView([1.043, -26.821, 13.465], 58.500, -17.565, .Models.Model)
    wait 10
    -- Now we pause the video to skip parts of the simulation
    F3DpauseVideo(true)
    wait 15
    -- Now we continue recording
    F3DpauseVideo(false)
    wait 10
    waituntil Source.Cont != void
    var MU = Source.Cont
    -- We now attach the camera to a part on the source
    F3DAttachCamera(MU)
    waituntil MU.Location = Conveyor2
    -- Detach the camera
    F3DAttachCamera(void)
    -- We set the camera to another position and direction
    F3DconfigureView([1.043, -26.821, 13.465], 58.500, -17.565, .Models.Model)

    wait 10
    -- Finish the video recording
    F3DfinishVideo
end
```

F3DpauseVideo [SimTalk]

Pauses the currently running video recording or continues playing it.

Type

Function

Syntax

```
F3DpauseVideo([StartPause:boolean:=true])
```

Parameter

The optional parameter *StartPause* of data type *boolean* sets if the recording will be paused (`true` or parameter not specified) or continued (`false`).

Default Value of the Parameter

The default value is true.

Example

```
F3DpauseVideo
```

See also

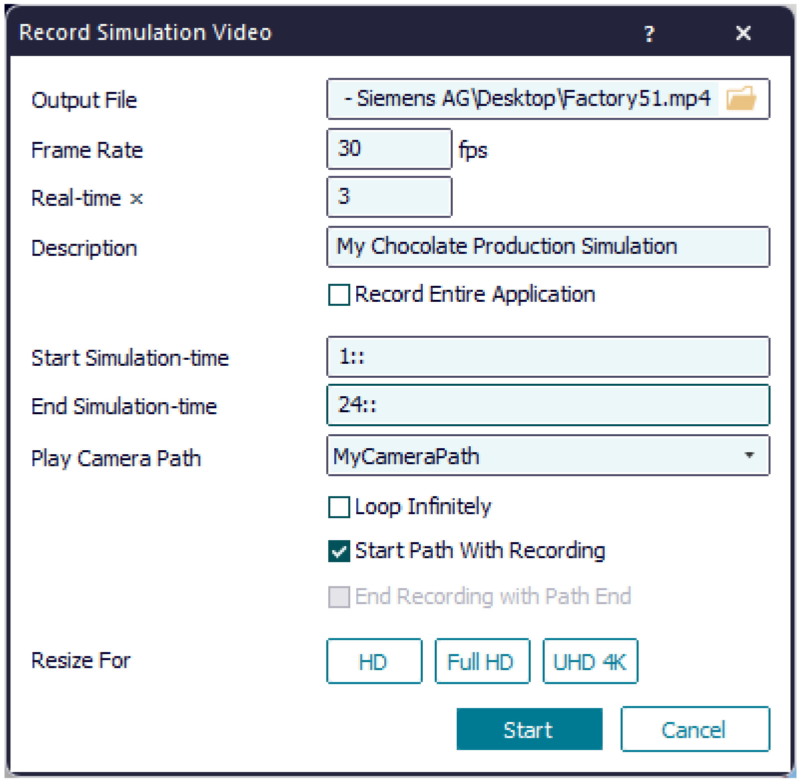
Video Ribbon Tab > **Pause/Resume Recording**

F3DrecordSimulationVideo [SimTalk]

Starts the simulation or resumes it and records a video of it.

Remarks

3D records the video with the current settings of the dialog **Record Simulation Video**.



Deactivate the safety setting **File > Model Settings > General > Prohibit Access to the Computer** to permit the function *F3DrecordSimulationVideo* to copy data from another folder and write it to the model folder or its sub-folders. The function cannot copy files from the model folder or its sub-folders.

Type

Function

Syntax

```
F3DrecordSimulationVideo(FilePath:string, ObjectOfInterest:object/3D
object, RealTimeScale:real[, FrameRate:integer:=25,
Description:string/void:="", RecordApplication:boolean:=false,
StartTime:time/void:=void, EndTime:time/void:=void, CameraPath:string:="",
LoopInfinitely:boolean:=false, StartPathWithRecording:boolean:=false,
EndRecordingWithPath:boolean:=false])
```

Parameters

You can specify the following parameters:

- The parameter *FilePath* of data type *string* designates the path to the video file that is going to be recorded.
- The parameter *ObjectOfInterest* of data type *object* or *3D object* designates the object from whose view the simulation is recorded. The object either has to be the *root Frame* of a simulation or be inserted into the *root Frame*. An object which does not have an *EventController*, for example a class, is not allowed.
- The parameter *RealTimeScale* of data type *real* designates the **real timescale** with which the video will be played later on.
- The optional parameter *FrameRate* of data type *integer* designates the frame rate for recording the video file. The default setting is 25 frames per second.

  Default Value of the Parameter

  The default value is 25.
- The optional parameter *Description* of data type *string* designates the descriptions that is shown on the first frame of the video.
  - Specify `void` to show no description on the first frame.
  - Specify an empty string `""` to use the default description, i.e., the file name of the model. The default setting is **""**.

  Default Value of the Parameter

  The default value is an empty string `""`.
- The optional parameter *RecordApplication* of data type *boolean* sets if the entire application window will be recorded (`true`) or only the window of the *ObjectOfInterest* (`false`).
  - Specify `false` to record a window of an *ObjectOfInterest*, provided such a window is open.
  - If no such window is open, *Plant Simulation* opens a suitable window.

  Default Value of the Parameter

  The default value is false.
- The optional parameter *StartTime* of data type *time* or *void* sets when recording starts.
  - Specify `void` or nothing to start recording when the simulation is started or continued.
  - If you specify a time, recording starts at this point in time.
  - If the point in time is located in the past, relative to the current simulation time, the simulation is going to be reset beforehand.

  Default Value of the Parameter

  The default value is void.
- The optional parameter *EndTime* of data type *time* or *void* sets when recording terminates.

- Specify `void` or nothing to never terminate recording by itself.
- If you specify a time, recording and simulation run up to this point in time of the simulation and then terminates. In addition, an **End Time** which you typed into the *EventController* will be deactivated for the duration of the recording.

Default Value of the Parameter

The default value is void.

- The optional parameter *CameraPath* of data type *string* designates the camera path that will be played during the simulation and the recording of the video. If you specify an empty string `""` or nothing no camera path is going to be played automatically.

> **Note**
>
> This parameter only applies to the *Frame*.

Default Value of the Parameter

The default value is an empty string `""`.

- The optional parameter *LoopInfinitely* of data type *boolean* sets if camera path is looped infinitely.
  - Specify `true` to play the path again from the start after reaching its end.
  - Specify `false` or nothing to make the view remain at it location when the end of the path is reached.

> **Note**
>
> This parameter only applies to the *Frame*.

Default Value of the Parameter

The default value is false.

- The optional parameter *StartPathWithRecording* of data type *boolean* sets if the camera path will be played when recording starts (`true`) or when the simulation starts/continues (false or nothing).

> **Note**
>
> This parameter only applies to the *Frame*.

Default Value of the Parameter

The default value is false.

- The optional parameter *EndRecordingWithPath* of data type *boolean* sets if simulation and video recording are also terminated when reaching the end of the camera path (`true`) or if simulation and video recording are to be continued after the end of the path has been reached (`false` nothing).

> **Note**
>
> This parameter only applies to the *Frame*.

Default Value of the Parameter

The default value is false.

Example

```
F3DrecordSimulationVideo("C:\Users\ha\Desktop\PlantSimulationVideo.avi", 6,
25, "")
```

See also

Prohibit Access to the Computer [model settings]

Start Simulation Recording

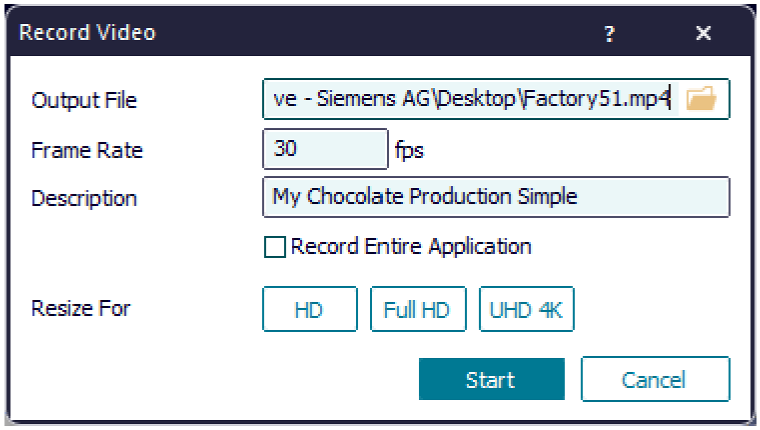
Video Ribbon Tab > Start Simulation Recording

F3DrecordVideo [SimTalk]

Records a video of *Plant Simulation* or of the active window.

Remarks

3D records the video with the current settings of the dialog **Record Video**.



If you deactivate the safety setting **File > Model Settings > General > Prohibit Access to the Computer**, the function *F3DrecordVideo* can copy data from another folder and write it to the model folder or its sub-folders. The function cannot copy files from the model folder or its sub-folders.

Type

Function

Syntax

```
F3DrecordVideo(FilePath:string[, FrameRate:integer:=25,
Description:string:="", RecordApplication:boolean:=false])
```

Parameters

You can specify the following parameters:

- The parameter *FilePath* of data type *string* designates the path to the video file that is going to be recorded. By default, the file is saved here **C:\Users\YourLoginName\Desktop**.
- The optional parameter *FrameRate* of data type *integer* designates the frame rate for recording the video file. The default setting is 25 frames per second.

  Default Value of the Parameter

  The default value is 25.

- The optional parameter *Description* of data type *string* designates the descriptions that is shown on the first frame of the video.
  - Specify `void` to show no description on the first frame.
  - Specify an empty string`""` to use the default description, i.e., the file name of the model.

  Default Value of the Parameter

  The default value is an empty string `""`.
- The optional parameter *RecordApplication* of data type *boolean* sets if the entire application window will be recorded (`true`) or only the 3D window of the *ObjectOfInterest* (`false`).

  Default Value of the Parameter

  The default value is false.

Example

```
F3DrecordVideo("C:\Users\ha\Desktop\PlantSimulationVideo.avi",
25, .Models.MyChocolatePlant)
```

See also

Prohibit Access to the Computer [model settings]

Video Ribbon Tab > **Start Simple Recording**